

3D Cadastre & Bhu-Aadhaar System

Comprehensive System Architecture and Technical Workflow

1. The Core Architecture

The system is orchestrated by a central engine (*main.py*) that connects a SQLite database to mathematical and GIS components.

- **Core Mathematical Engines:** Includes 3D floor slicing, ULPIN generation, collision detection, and cryptographic certification.
- **Presentation & GIS:** Utilizes PyDeck for 3D extrusion mapping and minimalist UI components for role-based workflows (Citizen, Surveyor, Registrar).

2. Core Mathematical Engines

A. Vertical Parcel Slicing

Traditional land records map 2D polygons (X, Y), which fails in high-rise scenarios where ownership varies along the Z -axis. This engine segments vertical space into standard units (default $3m$).

- **Superstructure:** $Z_{start} = Z_{base} + (f \times 3)$, $Z_{end} = Z_{base} + ((f + 1) \times 3)$
- **Subsurface:** $Z_{start} = -(b \times 3)$, $Z_{end} = -((b + 1) \times 3)$

B. 3D ULPIN Generator & Decoder

Extends the standard 14-digit Indian Bhu-Aadhaar ULPIN (State, District, Sub-District, Village/Ward, Plot ID) with an ISO 19152 compliant vertical suffix:

- **Above-ground:** e.g., -004 for Floor 4.
- **Underground:** e.g., $-U02$ for Basement 2.

Bidirectional parsing decodes the 18-character code to instantly locate the exact jurisdiction and floor level in 3D space.

C. 3D Subsurface Collision & Encroachment Engine

Detects when underground digs breach public utilities by calculating horizontal proximity (Haversine formula) and vertical interval overlap (ΔZ).

-  **Critical Direct Encroachment:** Distance $\leq (R_1 + R_2)$ AND $\text{Overlap}_Z > 0$.
-  **Safety Buffer Violation:** Distance $\leq (R_1 + R_2 + 20m)$.

D. Cryptographic Certificate & Dynamic QR Generator

Ensures tamper-proof verification using a SHA-256 digital signature hash generated from the ULPIN, Owner, Coordinates, Z-range, and Valuation. The system generates a dynamic QR code and an official PDF Title Deed.

3. GIS 3D Rendering & Viewports

Utilizes Deck.gl over CartoDB basemaps for dual-layer rendering:

- **ColumnLayer:** Renders 3D extruded polygonal prisms scaled by height and color-coded by asset class (e.g., Cyan for Commercial).
- **ScatterplotLayer:** Draws halo rings at ground level indicating surface and subsurface boundaries.

Adaptive viewport zooming automatically adjusts altitude, pitch, and radius based on whether the view is pan-India or zoomed in on a specific parcel.

4. Role-Based Workflow Operations

Persona	Primary Goal	System Actions
 Citizen / Homebuyer	Verify flat title before buying	Checks RERA status, encumbrances, previews 3D floor card with QR, downloads PDF.
 Government Surveyor	Measure spatial limits & verify permits	Calculates volume and FAR/FSI, runs automated 3D subsurface clash detection, ingests new parcels.
 Sub-Registrar	Execute legal conveyance of property	Simulates vertical deed transfer, calculates Stamp Duty, signs transaction onto cadastral ledger.

"Honorable judges, traditional 2D land systems like 7/12 extracts or Bhu-Naksha only map the Earth's surface. They cannot legally distinguish between 40 flat owners living on top of each other, nor can they prevent private basements from digging into multi-crore underground metro tunnels. Our solution—Bhu-Aadhaar 3D—upgrades India's cadastral system to volumetric 3D by extending 14-digit ULPINs into vertical airspace and subsurface identifiers, preventing urban infrastructure disasters via real-time collision detection, and empowering stakeholders with cryptographically sealed, QR-verifiable 3D Property Title Cards."